

LEDGER · 2026-08-28 · AGTECH / INDUSTRIAL BIOTECH

# Halophile-Derived Siderophores via Non-Sterile Hypersaline Fermentation

REJECTED

This concept did **not** clear the framework.  
The reasoning is published in full below.  
What the assessment found

|                       |                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Incumbent benchmark   | Sequestrene 138 Fe                                                                                                                          |
| Entry application     | Prevention and treatment of iron deficiency chlorosis in high-value orchard crops and vineyards in calcareous soils                         |
| Measured advantage    | 1.25x advantage on active stereoisomer content, but <1x regression on in-soil longevity requiring a double dose to match incumbent efficacy |
| Time to first revenue | 12 months                                                                                                                                   |
| Regulatory risk       | Moderate — qualification costs reported (\$50,000 estimated)                                                                                |

## Why it failed

- Voided by: Economics undisclosed, Negative control / sub-2x

## Assessment

The venture explicitly concedes failure across multiple critical criteria. Startup CapEx of \$400,000 exceeds the \$250,000 limit, gross margins are negative (-262.8%) due to extensive downstream water removal costs from low-titer broth, and the product fails to achieve a 2x performance superiority over Fe-EDDHA due to rapid soil biodegradation.

### Also flagged

- no\_equipment\_list
- no\_opex

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

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Assessed 2026-08-28 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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